LOCALISATION COMMUTES WITH TAKING THE QUOTIENT BY A SUBMODULE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Exactness of localisation. Invoked implicitly between the two displayed sequences in the counted proof (lines 1579-1587). Carries the short exact sequence of modules to the short exact sequence of localisations.
- The quotient as the cokernel of an inclusion. Reads the last term of the localised sequence at proof lines 1585-1588. Identifies S^{-1}(M/N) with the quotient of S^{-1}M by S^{-1}N.

Lemma 0.1. *Localization respects quotients, i.e. if N is a submodule of M, then $S^{-1}(M/N) \simeq (S^{-1}M)/(S^{-1}N)$.*

Proof. From the exact sequence

$$0 \longrightarrow N \longrightarrow M \longrightarrow M/N \longrightarrow 0$$

we have

$$0 \longrightarrow S^{-1}N \longrightarrow S^{-1}M \longrightarrow S^{-1}(M/N) \longrightarrow 0$$

The corollary then follows. □

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